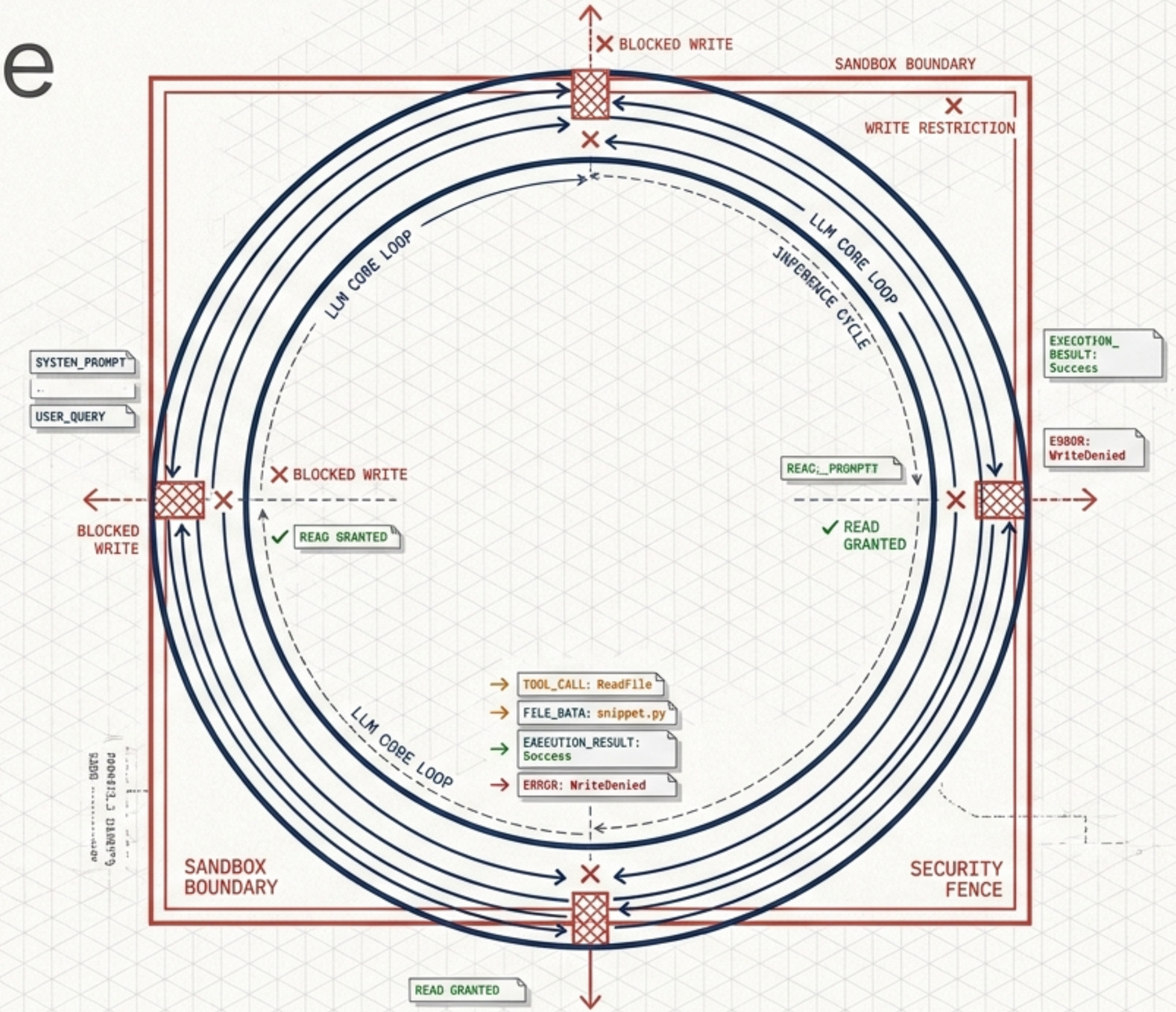
Demystifying the Coding Agent

An architectural deconstruction of an LLM in a loop.

A scene-by-scene explainer tracing how AI actually interacts with code, tools, and the filesystem.



Gemini Notebook

[LLM] ⟳ [Tools] + [Fence]

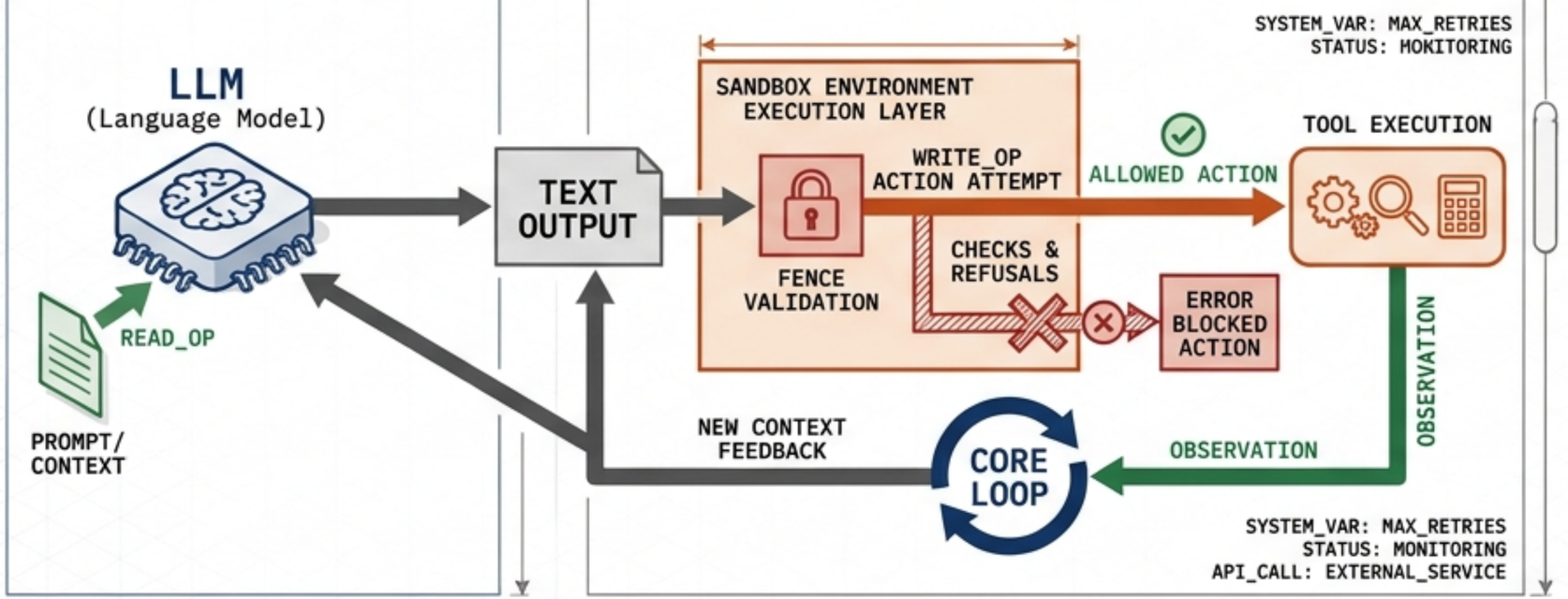
An agent is just an LLM in a loop with tools and a fence

The Engine

The model has no hands. It only writes text.

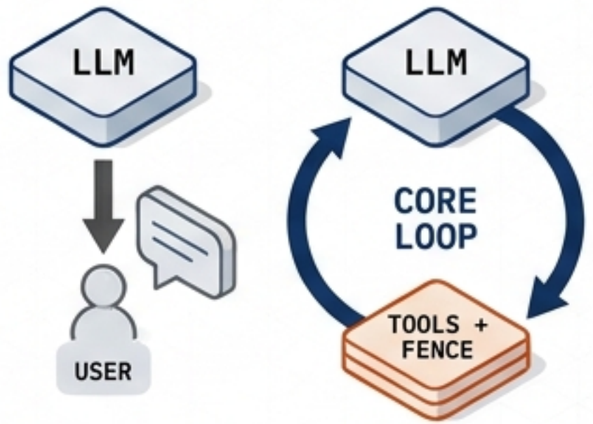
The Harness

The surrounding program acts, checks, and refuses.



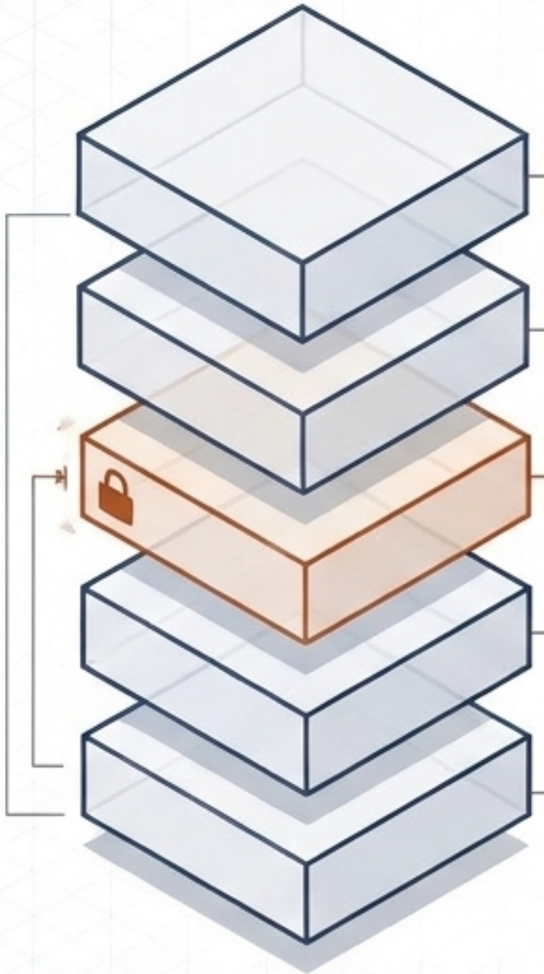
The Difference

Remove the loop, and you have a chatbot that can only talk. Keep it, and the model can look, act, observe, and try again.



Gemini Notebook

Building a strictly controlled world before the model wakes up



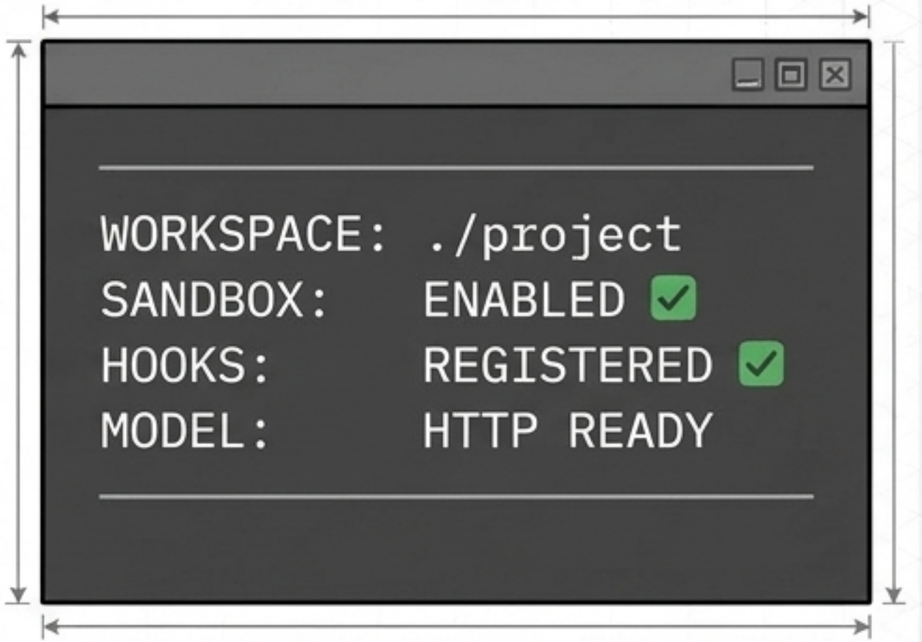
Model - Connects the HTTP chat function (live or demo).

Hooks - Registers the lifecycle observation moments.

Sandbox - Erects the fence against forbidden commands.

Workspace - Defines where reads start and bounds for writes.

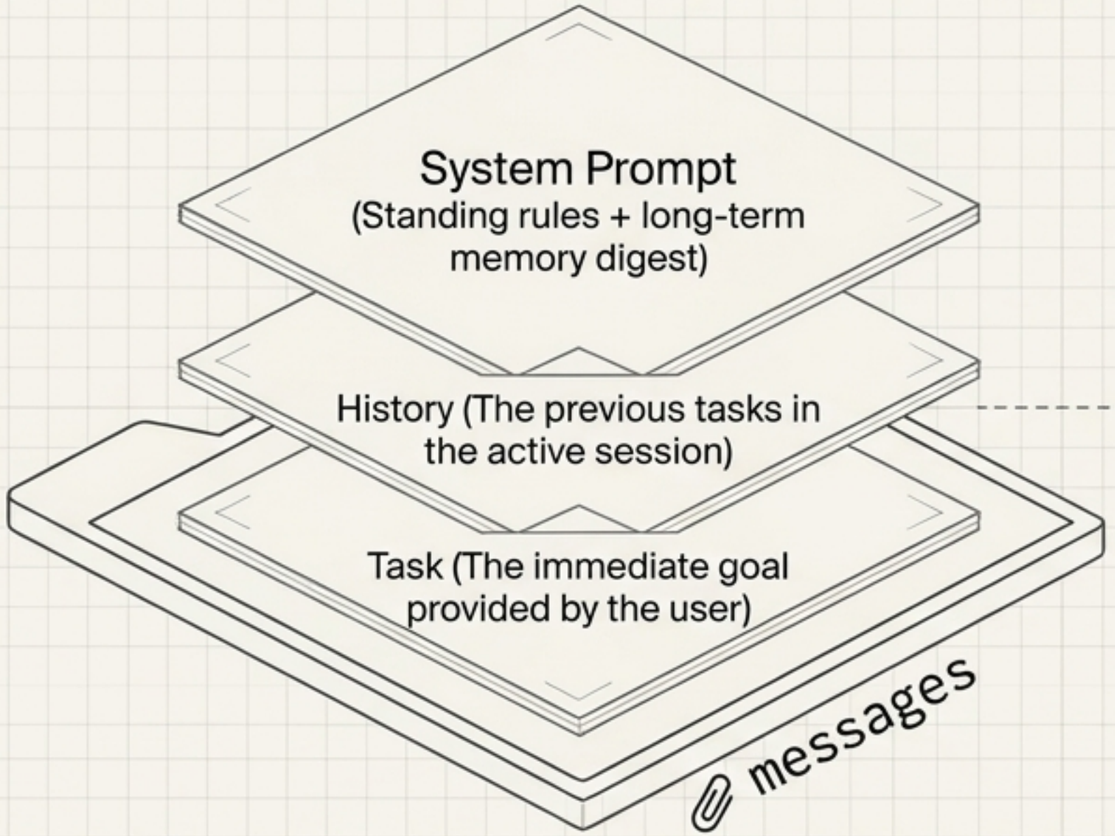
Config - Finds .env, merges with shell, sets key and budgets.



The banner printed by the CLI is a receipt proving exactly which world this run lives in.

Gemini Notebook

The model has absolutely no hidden memory



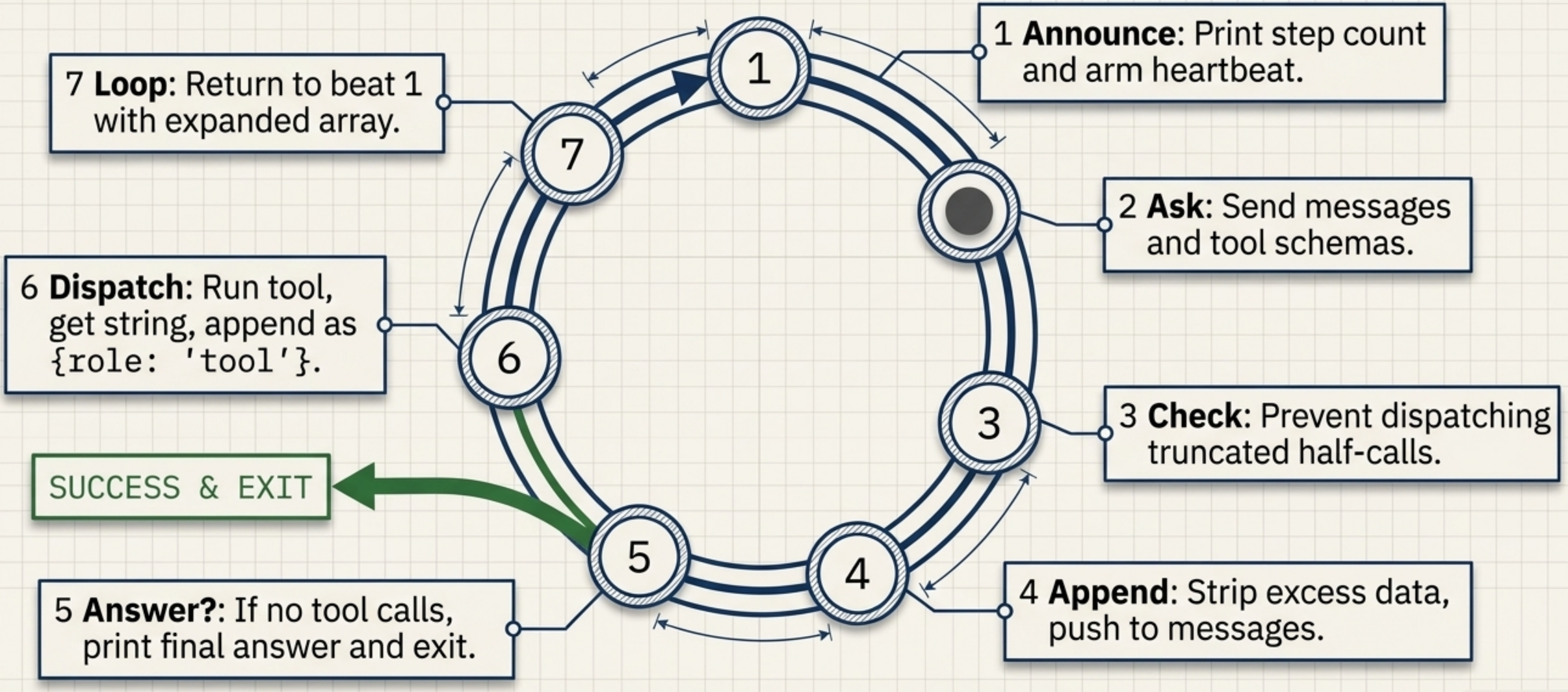
Tool Schemas

```
function: generate_image
(prompt, parameters)
function: search_database
(query, filters)
function: ...
function: calculate_metrics
(data_source, metric_type)
function: ...
function: ...
```

If something is not in this messages array, the system prompt, or the tool schemas, it simply does not exist for this step.

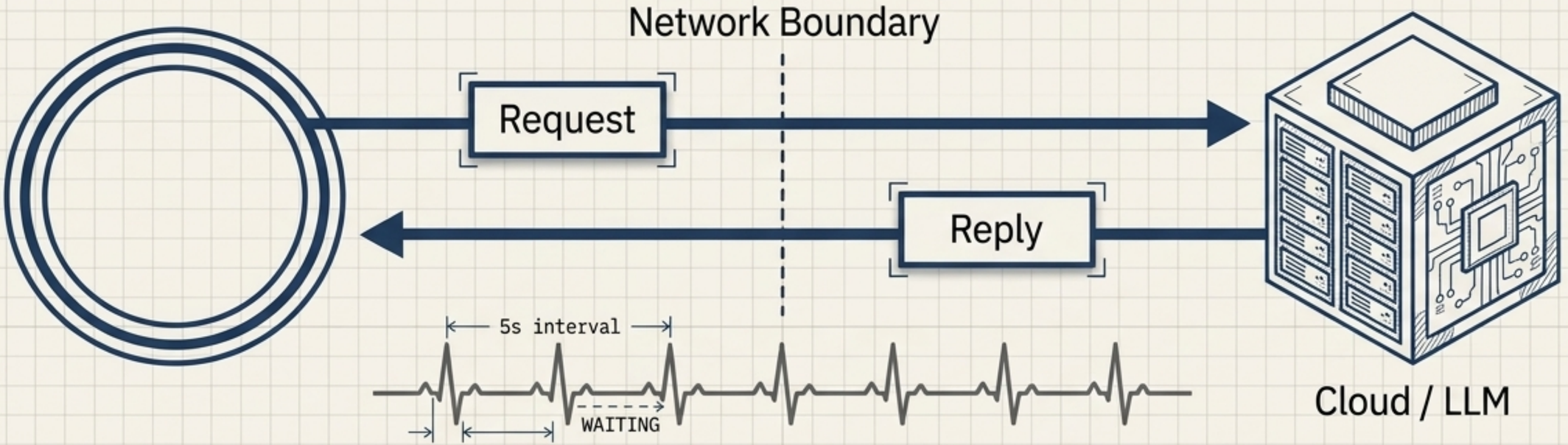
Gemini Notebook

One task executes as a strict seven-beat loop



Gemini Notebook

Every model request is sanitized, bounded, and monitored



Bounded Request

Hard limit of `2,048` **tokens** keeps replies small; a `timeout` prevents stalled providers from hanging the run.

The Heartbeat

Prints "`still waiting`" every 5 seconds so a slow network call never looks like a frozen program.

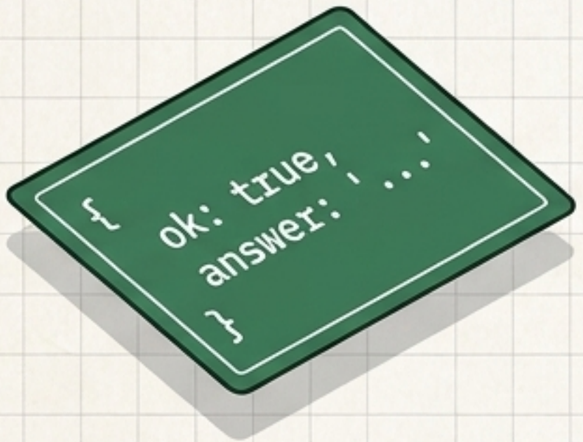
Sanitized Reply

Private reasoning fields are actively dropped. Only `content` and `tool_calls` are appended to history, preventing token bloat.

Gemini Notebook

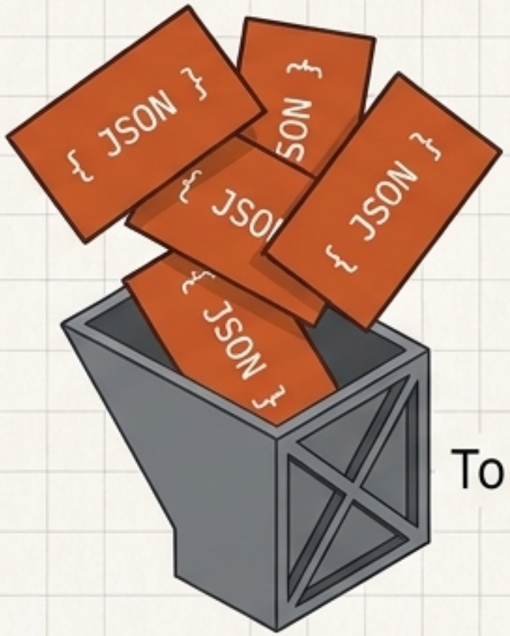
The model returns either a final answer or untrusted actions

Heads



Answer: No tool calls present. The loop prints the content and immediately returns.

Tails



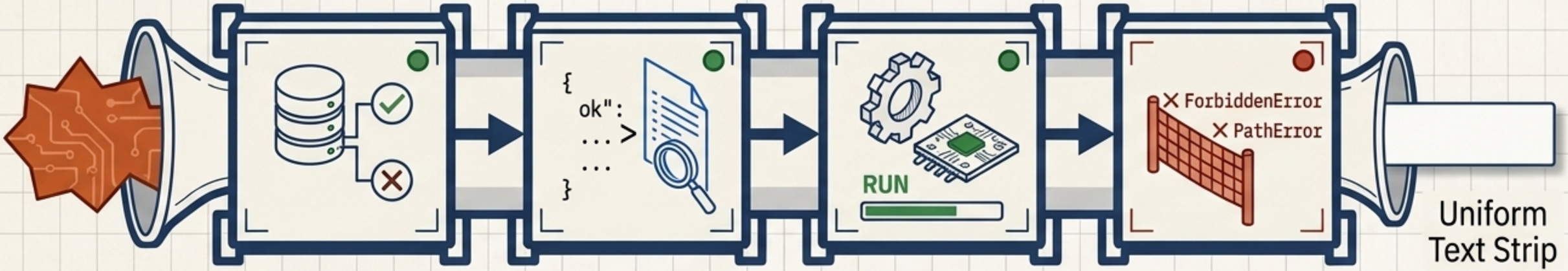
To Dispatcher

Action: The arguments are raw, unverified JSON strings. They may name non-existent tools, use wrong types, or target forbidden paths.

Critical Protocol: Every single tool call must get a matching tool message with the same ID appended to the history, even if it fails, or the provider API will reject the next step.

Gemini Notebook

The dispatcher translates untrusted intent into safe reality



Untrusted ⚠ JSON

Lookup: Rejects unknown tool names.

Validate: Parses JSON and checks types against schema.

Run: Executes the handler.

Catch: Traps all `ForbiddenError` or `PathError` exceptions.

The Golden Rule: The dispatcher always returns a `string`. A thrown `error` crashes a program, but a returned text `string` becomes an "`observation`" the model can read and recover from on the next `loop`.

Gemini Notebook

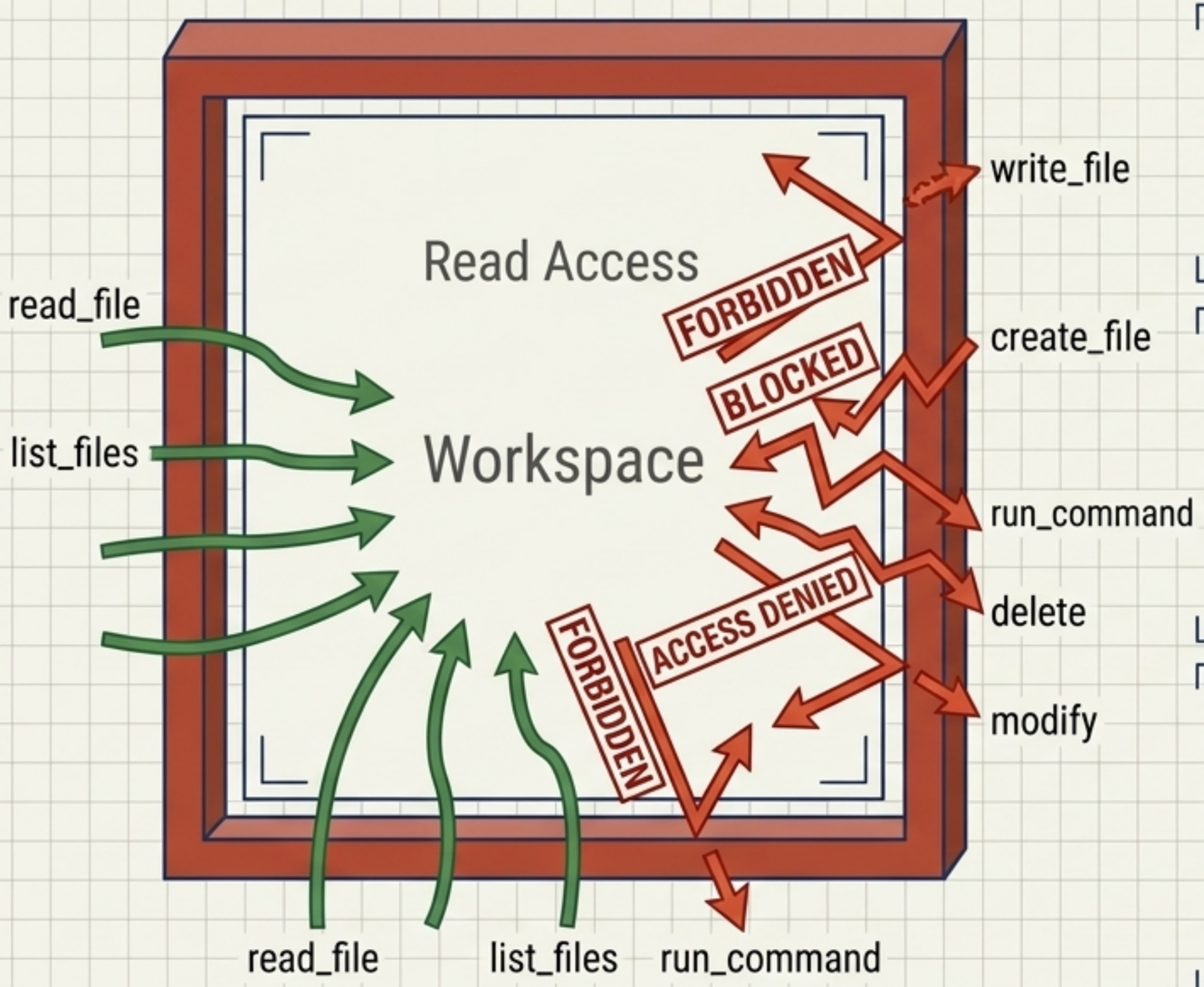
Eight specific tools with two deliberate asymmetries

	Read	Write	Execute
Open	 list_files, read_file, recall. A blind agent is useless.		
Fenced & Strict Policy		 create, write, edit, remember. A free agent is dangerous.	 run_command. Policy-gated and scanned.

The Output Asymmetry Actions return small summary strings (e.g., ‘1 replacement at line 2’), not entire files. Re-sending the whole file wastes the context window.

Gemini Notebook

Best-effort guardrails bound the model's autonomy



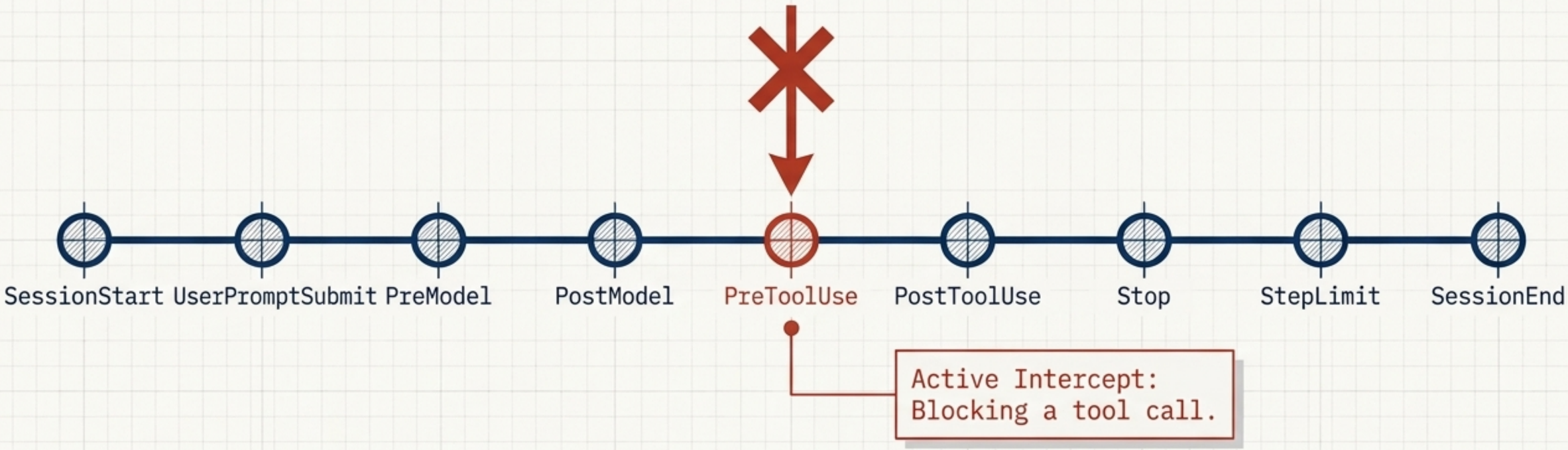
The Sandbox
Blocks absolute paths, out-of-bounds traversals, and home/system folder names.

The Command Policy
Deny-lists dangerous shell commands (`fork bombs`, `shutdowns`) and scans for shell-write targets escaping the workspace.

Environment Allow-list
Child processes only receive 13 explicitly safe environment variables. **API keys** are hidden to prevent accidental echoing.

Gemini Notebook

Nine named moments allow observation without breaking the loop



The Hook Architecture

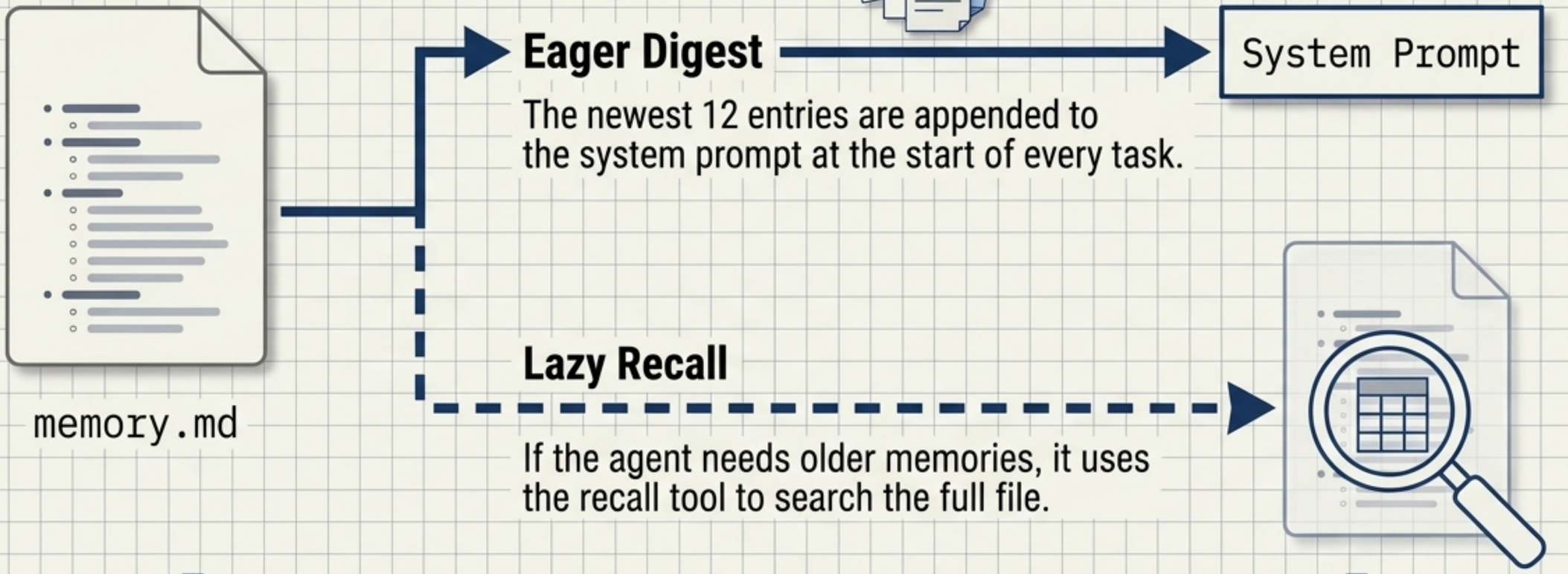
A fixed loop is safe but rigid. Hooks let external code interact at exact lifecycle moments without breaking the loop's invariants.

Hook Capabilities

Hooks can Observe (measure tokens, log notes), Rewrite (alter task or tool results), or Block (refuse a call with a reason).

Gemini Notebook

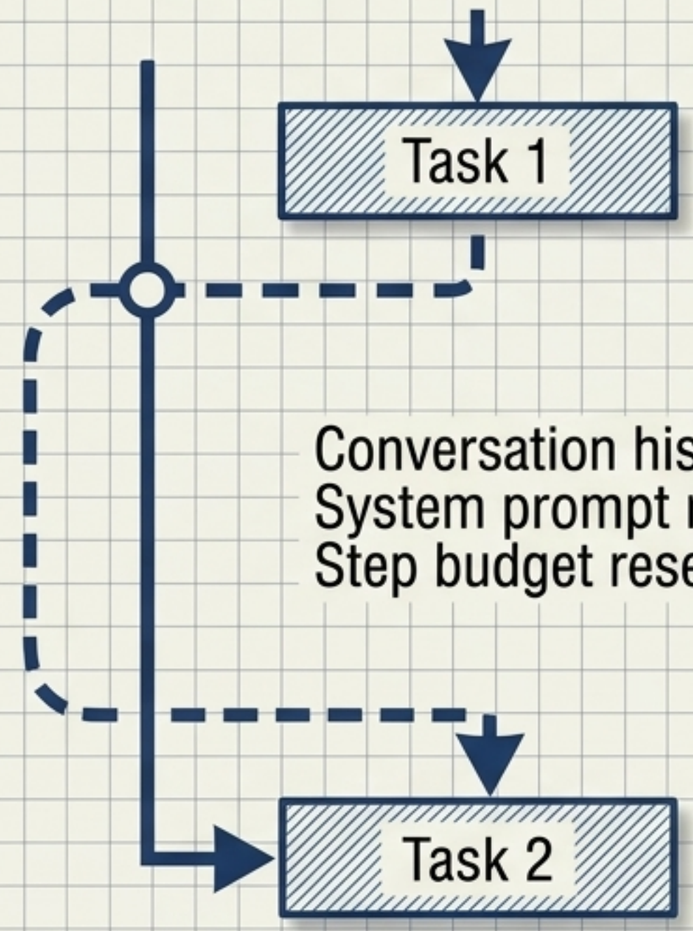
Durable facts survive the process in a plain text file



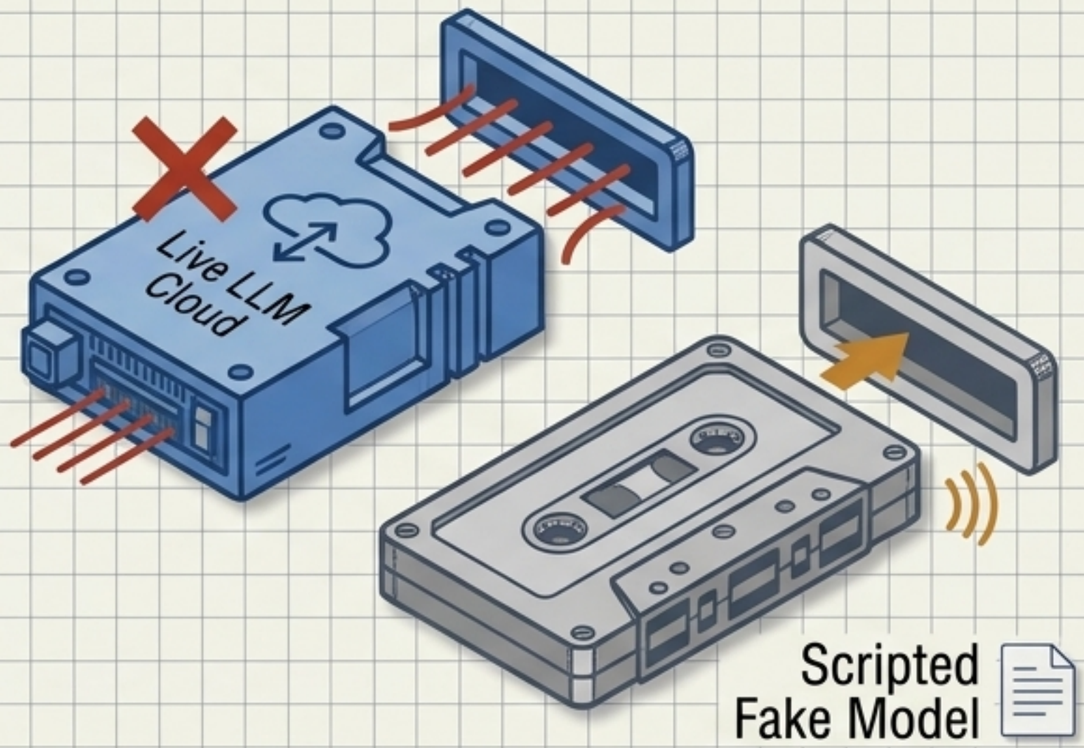
The Concept:
The `messages` array dies with the process; `memory.md` persists.
No vector databases, no magic—just plain markdown with bullets and dates.

Gemini Notebook

Sessions carry context while demo mode tests the harness for free



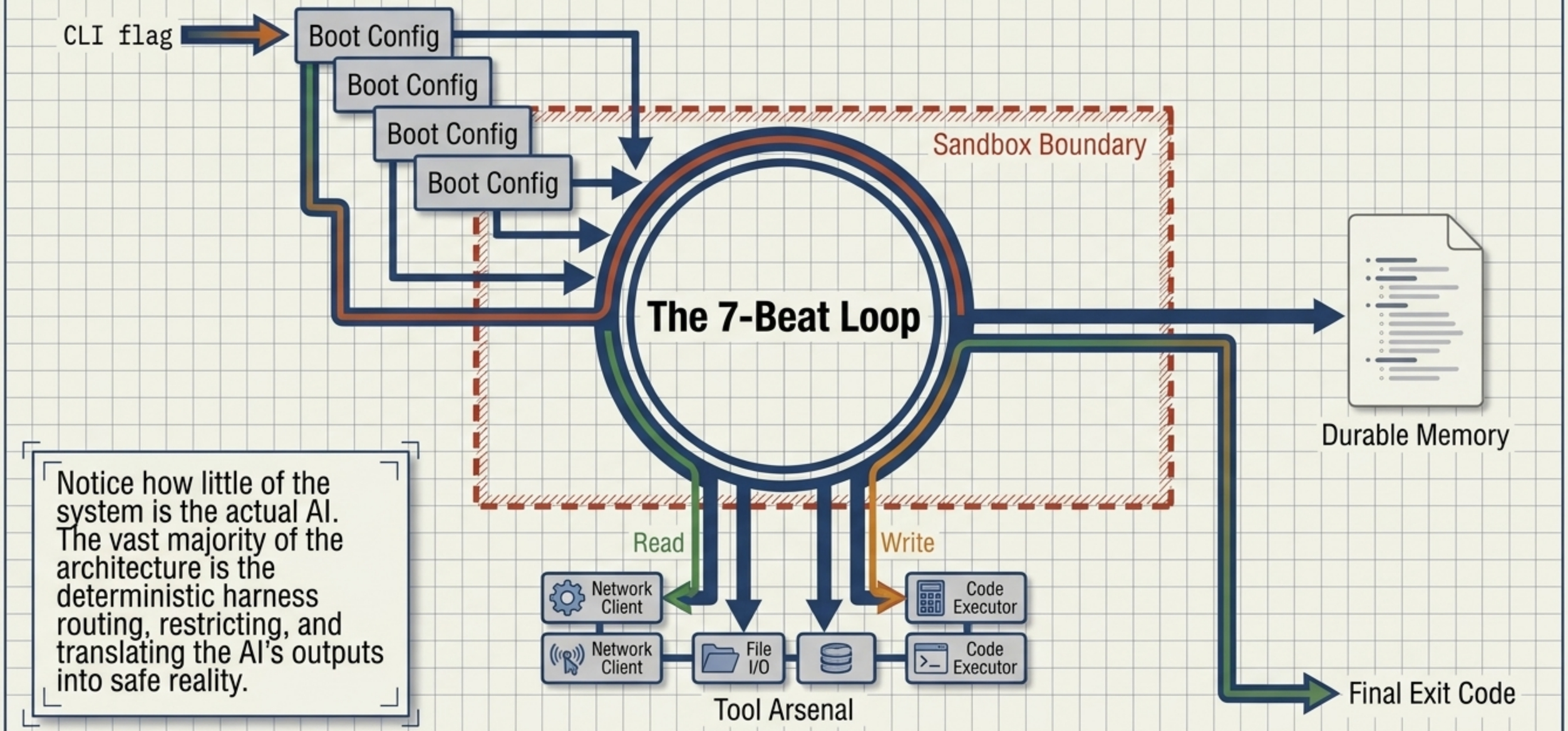
Conversation history carries over.
System prompt rebuilds.
Step budget resets to 1.



Replaces only the LLM with a fake script. The loop, tools, and fences remain 100% real, proving the guardrails hold without spending API credits.

Gemini Notebook

The entire architecture exists merely to manage the AI's interactions



Notice how little of the system is the actual AI. The vast majority of the architecture is the deterministic harness routing, restricting, and translating the AI's outputs into safe reality.

Gemini Notebook

The absolute laws of the coding agent

1. **The loop is the agent.**
2. **The messages array is the absolute state.**
3. **Errors are observations, not crashes.**
4. **The model is deeply untrusted.**
5. **Memory is just a readable text file.**

A model asks, a dispatcher acts, a sandbox refuses, a loop repeats, and a file remembers — until the model answers.

Gemini Notebook